REMARKS ON TOPOLOGICAL STRINGS

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ABSTRACT. We study a local mixed A/B topological string model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$. The open theory reduces to a one-dimensional gauged first-order matrix model, while the closed local model is the cotangent BF theory for Hamiltonian vector fields on the holomorphic symplectic surface. We construct the classical bulk-to-boundary map by evaluating closed Hamiltonians on the matrix normal coordinates of the brane and prove that it intertwines the closed Poisson bracket with the open trace bracket. In the stable large-N limit this identifies the reduced Hamiltonian sector of the classical brane operators with the PBW limit of the Koszul dual of the closed Chevalley-Eilenberg algebra.

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1. Introduction

1.1. **Dreams.** One of the prevalent themes at play in both contemporary theoretical physics and mathematics is that of *duality*. In particular, we will adopt a form of *Koszul Duality* as a guiding light in our quest to understand some of the salient characteristics of **twisted M-Theory**.

Remark 1.1. This introduction serves a motivational purpose and consequently de-emphasises rigour in favour of intuition; the precise statements appear in the body.

Let X be a holomorphic symplectic manifold. We will consider a topological twist of M-theory on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$, for some point $p \in X$.

In this scenario one expects to be able to construct two algebras:

- The algebra $\mathcal{A}_N$ of operators on the stack of N branes supported on $\mathbb{R} \times p$.
- The algebra $\mathcal{B}$ of local operators of the gravitational theory on $\mathbb{R} \times X$. Operator product expansions in the direction of $\mathbb{R}$ equip $\mathcal{B}$ with the structure of an associative algebra.

Twisted holography then leads us to expect:

Conjecture 1.2. As $N \to \infty$, $\mathcal{A}_N$ is the Koszul dual of $\mathcal{B}$.

Remark 1.3. One should be able to exploit Witten's proposal [4] for holographic calculations of the OPE.

These notes isolate one local mixed A/B example and verify the classical stable operator-algebra match. The quantum discussion is restricted to the canonical first-order bracket and the formal odd-power Moyal targets; the full closed-string Feynman integral calculation is not claimed here.

1.2. **The local theorem.** Let

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1, \qquad \mathfrak{h} = \bar{A}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}[1],$$

where $\mathfrak{h}$ carries the Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g,$$

and $\mathfrak{h}[1]$ is the adjoint module. Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field, while on the open side $\mathrm{Tr}(1) = N$ is a scalar parameter rather than a nontrivial operator.

Theorem 1.4 (Local stable Hamiltonian-sector Koszul duality)**.** *In the formal local mixed A/B model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$, the stable large-N reduced open Hamiltonian-sector operator algebra is*

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}[1]\big).$$

The closed local operator algebra is the continuous Chevalley-Eilenberg algebra

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^{\vee}[-1]\big).$$

Its Koszul dual is the completed graded enveloping algebra

$$\big(\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}}\big)^{!} \simeq \widehat{U(\mathfrak{g})}.$$

The Rees algebra $U_\hbar(\mathfrak{g})$ has special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}[1]\big),$$

which identifies the PBW classical limit of the closed Koszul dual with the open stable Hamiltonian-sector algebra. The identification is induced by the bulk-to-boundary evaluation map

$$\partial_{\mathrm{bb}}(f) = \mathrm{Tr}\, f(\phi_1, \phi_2),$$

which sends a closed Hamiltonian to the corresponding brane Hamiltonian and intertwines Poisson brackets. On monomials,

$$z_1^k z_2^l \longleftrightarrow \mathrm{Tr}(\phi_1^k \phi_2^l), \qquad k + l > 0,$$

the first bracket is

$$\{\mathrm{Tr}(\phi_1^k \phi_2^l), \mathrm{Tr}(\phi_1^m \phi_2^n)\} = (kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1}),$$

matching the Poisson bracket on $\bar{A}$.

1.3. **Status of assertions.** The theorem above is what is proved in these notes. The proof consists of the affine Ext calculation, the open BV reduction, the stable invariant-theory calculation of the Hamiltonian brane sector, the formal Hamiltonian BF model for the closed side, the construction of the bulk-to-boundary Hamiltonian evaluation map, the CE/Koszul-duality computation, and the first bracket calculation.

The Moyal $\hbar^3$ formula below is a formal deformation-quantization target checked by an exact script. It is not presented as the value of a closed-string Feynman integral. The ambient twisted-holography

and M-theory discussion is motivation for this local theorem, not an input to its proof.

1.4. **Conventions for the local calculation.** Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. Thus the copy $\bar{A}[1]$ in $\bar{A} \oplus \bar{A}[1]$ is the odd antifield copy. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space V.

Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and then passing to local operators in the interior. The term *stable large-N* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$; no finite-N trace identity is being suppressed inside a fixed finite matrix algebra. The scalar trace $\mathrm{Tr}(1) = N$ is treated as a parameter and is removed from the reduced operator algebra.

On the open side we compute the Hamiltonian sector: the subalgebra of reduced BV cohomology generated by the ghost-zero traces and their first Koszul descendants, i.e. by single traces with no ψ and by single traces with one ψ. The full stable BV invariant cohomology may contain higher single-trace antifield sectors; those are not identified in this paper and are not used in the Koszul-duality comparison.

1.5. **The concrete dictionary.** The comparison is the following explicit dictionary. For $k, l \geq 0$ and $k + l > 0$, set

$$H_{k,l} = z_1^k z_2^l \in \bar{A}, \qquad E_{k,l} = z_1^k z_2^l[1] \in \bar{A}[1].$$

On the brane side set

$$O_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l),$$

and let $\Psi_{k,l}$ be the symmetrised trace obtained by inserting one ψ into the word with k copies of ϕ_1 and l copies of ϕ_2; the odd matrix ψ is the Koszul variable for the moment-map equation, introduced in Construction 2.15. The theorem proves that the assignment

$$H_{k,l} \longmapsto O_{k,l}, \qquad E_{k,l} \longmapsto \Psi_{k,l}$$

extends to an isomorphism

$$\mathbf{S}(\bar{A} \oplus \bar{A}[1]) \xrightarrow{\sim} \mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}}$$

in the Hamiltonian sector. On the closed side the same symbols $H_{k,l}, E_{k,l}$ form the PBW generators of the Rees specialization of $U_\hbar(\bar{A} \ltimes \bar{A}[1])$.

Thus every later step has a concrete target: construct the symbols $O_{k,l}, \Psi_{k,l}$ from the open fields, construct $H_{k,l}, E_{k,l}$ from Hamiltonians and cotangent Hamiltonians, prove that commutators of words reduce to commutative monomials, and check that the bracket coefficient is $(kn - lm)$.

1.6. **String Field Theory and Koszul Duality.** We expect Koszul duality for string field theories to be reflected in stable trace invariants and cyclic homology. The algebraic template is the Loday-Quillen-Tsygan theorem [2]tsygan:

Theorem 1.5 (Loday-Quillen-Tsygan)**.** *Let A be a unital associative algebra over a field of characteristic zero. The stable Chevalley-Eilenberg homology of $\mathfrak{gl}_\infty(A)$ is the free graded-commutative algebra generated by the shifted cyclic homology of A:*

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A)\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Thus stable matrix invariants are controlled by cyclic words. In the particular reduced model below, BRST exactness of $[\phi_1, \phi_2]$ turns cyclic words into ordinary commutative monomials, and the surviving single-trace generators are represented by $\mathrm{Tr}(\phi_1^k \phi_2^l)$ with $k + l > 0$.

The calculation below is the corresponding classical shadow: closed Hamiltonian modes match stable single-trace generators, and the first nontrivial bracket agrees with the Poisson bracket.

1.7. **Flavours of Topological Strings.** Topological strings manifest in various flavours. Common examples include:

- The **A-model**, where branes are objects of a Fukaya category: Lagrangians equipped with the grading, spin or Pin, local-system, and unobstructedness data required to define Floer theory.
- The **B-model**, where branes are objects of $D^b\mathrm{Coh}(X)$, or of $\mathrm{Perf}(X)$ in the affine local model.
- **Mixed A-B models**, e.g. on $X \times X'$ with branes given by $L \times Z$ for $L \rightarrowtail X$ Lagrangian and $Z \rightarrowtail X'$ holomorphic.

Remark 1.6. The A-model is only interesting with world-sheet instantons taken into account.

Regardless of the context we will work in, we need to understand:

- The open string theory, which is the theory on a brane, and
- The closed string field theory, of a gravitational nature.

1.8. **Open Strings.** If $\mathcal{C}$ is a cyclic Calabi-Yau A_∞ category and $\mathcal{F} \in \operatorname{Ob}\mathcal{C}$, set $A = \operatorname{RHom}(\mathcal{F}, \mathcal{F})$. The open-string states are A. The BV fields are[1]

$$A[1].$$

The Calabi-Yau structure supplies a cyclic trace $\operatorname{Tr} : A \to \mathbb{C}[-d]$. For a cyclic A_∞ algebra with operations m_n, the formal Chern-Simons action is

$$S(a) = \sum_{n \geq 1} \frac{1}{n+1} \operatorname{Tr}\big(a\, m_n(a, \ldots, a)\big), \qquad a \in A[1].$$

This is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [5].

2. The Local Model

2.1. **Setup.** Let $X = \mathbb{R}^2_A \times \mathbb{C}^2_B$. We use the exact A-brane given by the standard line $\mathbb{R} \subset \mathbb{R}^2_A$ with trivial brane data and the B-brane $\mathcal{O}_p \in \operatorname{Perf}(\mathbb{C}^2_B)$. Consider the product brane

$$\mathbb{R} \times p \rightarrowtail \mathbb{R}^2_A \times \mathbb{C}^2_B$$

for some point $p \in \mathbb{C}^2_B$. The A-model lives on $\mathbb{R}^2_A$ (equipped with its standard symplectic structure) and the B-model lives on $\mathbb{C}^2_B$ (equipped with its standard holomorphic-symplectic structure $\omega = dz_1 \wedge dz_2$).

2.2. **Open String Field States.** The open string states are the tensor product of A-model and B-model states.

- *A-model factor.* In the exact self-Floer model for $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, a small Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex of $\mathbb{R}$. Hence $HF^*(\mathbb{R}, \mathbb{R}) \cong H^*(\mathbb{R}; \mathbb{C})$, and the de Rham complex $\Omega^*(\mathbb{R})$ is the chain-level model used below.

[1] If we produce a category, the resulting field theory is defined on a point. If instead we produce a sheaf of categories, we get a field theory on the support of this sheaf, i.e. a field theory on the given brane.

- *B-model factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states
$$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$
computed in Proposition 2.1.

Proposition 2.1 (Affine skyscraper self-Ext)**.** *Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is*
$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$
where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in the polynomial ring R, so the affine Koszul complex
$$K = \big(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\big)$$
is a free resolution of k, with
$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$
Applying $\mathrm{Hom}_R(-, k)$ gives
$$0 \to k \xrightarrow{0} k\,\varepsilon_1 \oplus k\,\varepsilon_2 \xrightarrow{0} k\,\varepsilon_{12} \to 0,$$
because z_1 and z_2 act by zero on k. Hence the graded vector space of Ext groups is k in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends e_i to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

2.3. **Open String Field Theory.** The total space of fields (in the Batalin-Vilkovisky formalism) is
$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$
with action
$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\,\mathrm{Tr}(\alpha\, d\alpha) + \frac{1}{3}\,\mathrm{Tr}\,\alpha^3.$$
Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

Construction 2.2 (The open field from the Ext basis)**.** Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are exactly

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and they appear as

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The Berezin integral kills every monomial except the coefficient of ε_{12}, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

Remark 2.3. The shift $[1]$ acts on the total cohomological degree: a form of de Rham degree p and Grassmann degree q has total degree $p+q$, which after the shift sits in ghost number $p+q-1$. In particular, the 1-form component of α (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to ε_i (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

2.3.1. *De-BV-fying the open string field theory.* The ghost-number-zero part of the BV field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2,$$

where $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$ and $\phi_i \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$.

Lemma 2.4. *With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to α_0 is, up to a boundary term on $\mathbb{R}$,*

$$S(\alpha_0) = \int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive the integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In the quadratic term this leaves

$$\frac{1}{2}\int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 - \phi_2\, d\phi_1\big),$$

where the sign is the Koszul sign from interchanging the two odd generators. Since $d\operatorname{Tr}(\phi_1\phi_2) = \operatorname{Tr}(d\phi_1\,\phi_2 + \phi_1\, d\phi_2)$, the second summand differs from $\operatorname{Tr}(\phi_1\, d\phi_2)$ by an exact one-form. In the cubic term the six orderings of $A, \phi_1\varepsilon_1, \phi_2\varepsilon_2$ reduce by cyclicity of the trace

and the odd relation $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$ to $\mathrm{Tr}(A\phi_1\phi_2 - A\phi_2\phi_1)$. This is $\mathrm{Tr}(A[\phi_1, \phi_2])$. □

Starting from the BV master action and integrating out the antifields (i.e. passing to the physical field content by restricting to ghost-number-zero fields), we find a quantum-mechanical system with a connection form

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

two adjoint scalars

$$\phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and action

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

This action is invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon],$$

for $\varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$. Indeed the variation of $\mathrm{Tr}(\phi_1 d\phi_2)$ containing $d\varepsilon$ is $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, and this cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

Remark 2.5. One can also proceed directly within the Calabi-Yau category: take a given object, tensor with $\mathbb{C}^N$, and pass to endomorphisms. More generally, one should work with the cyclic cohomology of the category; the 3-point function is computable via the cup-product in cyclic cohomology together with the trace, while higher n-point functions require additional structure.

Remark 2.6. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$, symplectic form $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$, and GL_N moment map $\mu = [\phi_1, \phi_2]$. The classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

Remark 2.7. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over T^2 gives the same action; the odd variables ε_i are the algebraic avatars of the degree-one cohomology classes η_i.

2.4. **Closed String States.** Similarly, the closed string state space is of the form

$$\text{A-model closed states} \otimes \text{B-model closed states.}$$

For the A-model on $\mathbb{R}^2$, we take the de Rham cohomology complex $\Omega^*(\mathbb{R}^2)$.

For the B-model on $\mathbb{C}^2$, we use the local holomorphic-symplectic Kodaira-Spencer model obtained from Hamiltonian holomorphic vector fields. It is BCOV-like in its use of polyvectors and the divergence operator, but it is not the Calabi-Yau threefold BCOV theory with one sector simply deleted.

Remark 2.8. The original BCOV analysis [1] is written for Calabi-Yau threefolds. On a holomorphic symplectic surface the natural local cotangent theory is the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the holomorphic volume form supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where ∂_{z_i} are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \to \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\text{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1} a_1 + \partial_{z_2} a_2),$$

while ∂_Ω vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. In this model the small closed-string state space is the ∂_Ω-closed part of polyvectors. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification $\mathbf{Ker}\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 2.9 identifies these with df, uniquely modulo constants.

- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra
$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$
for nontrivial Hamiltonian fields.

Lemma 2.9 (Polynomial Poincare primitive)**.** *Let*
$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$
be a polynomial one-form on $\mathbb{C}^2$*. If* $\partial_{z_2} a = \partial_{z_1} b$*, then there is a polynomial* $f \in \mathbb{C}[z_1, z_2]$*, unique modulo constants, such that* $\theta = df$*.*

Proof. Define
$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$
Since a and b are polynomials, the integral is just division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so
$$\begin{aligned}\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\ &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\ &= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).\end{aligned}$$
The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

Remark 2.10. A global analytic theory on noncompact $\mathbb{C}^2$ requires boundary conditions, a gauge slice, and a choice of Green kernel. The computations below use only the formal local algebra of polynomial Hamiltonians and do not require a global propagator.

For the closed string field theory we replace $\mathbf{Ker}\partial$ (on poly-1-vectors) by the Dolbeault-valued Hamiltonian vector fields:
$$\Omega^{0,*}(\mathbb{C}^2)\text{-valued Hamiltonian vector fields,}$$
i.e. sections of $\Omega^{0,*}(\mathbb{C}^2)$ identified with Hamiltonian vector fields via the symplectic form $\omega = dz_1 \wedge dz_2$. Concretely, a section $f \in \Omega^{0,*}(\mathbb{C}^2)$ corresponds to the Hamiltonian vector field $X_f = \{f, -\}_\omega$, where
$$\{f, g\}_\omega = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

Remark 2.11. The replacement is the elementary statement

$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$

The first isomorphism uses the holomorphic symplectic form and the second is the polynomial Poincare lemma on affine space. Constants are removed because $X_f = 0$ exactly when f is constant.

2.5. **Closed String Field Theory.**

2.5.1. *The Action Functional.* Let

$$D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

Write

$$\alpha \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \otimes \bar{A}[1]$$

for the Hamiltonian connection field and

$$\beta \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \otimes \bar{A}^\vee[2]$$

for its cotangent field. The bracket on form-valued Hamiltonians is obtained by wedging the form parts and taking the Poisson bracket of the Hamiltonian parts. The action is

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1\, dz_2.$$

The pairing is the evaluation pairing between $\bar{A}^\vee$ and $\bar{A}$ (or, in a formal residue model, the corresponding continuous residue pairing). For ghost-number-zero fields, α has total real/Dolbeault form degree 1 and β has total degree 2. Hence $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total real/Dolbeault degree 4; multiplying by $dz_1 \wedge dz_2$ gives a top real degree 6 form on $\mathbb{R}^2 \times \mathbb{C}^2$.

Remark 2.12. Before, the quantum mechanical system lived on a line (the extent of the brane), while this gravitational theory occupies all of space-time.

2.5.2. *De-BV-fying the closed string field theory.* This theory is closely related to a 4-dimensional BF theory: β plays the role of the B-field (Lagrange multiplier) and the equations of motion of β enforce $F_\alpha = 0$, where

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\}$$

is the "curvature" of α (viewed as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$).

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of α and β are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar{z}_j}\, d\bar{z}_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar{z}_j}\, dx_i\, d\bar{z}_j + \beta_{\bar{z}_1 \bar{z}_2}\, d\bar{z}_1\, d\bar{z}_2.$$

Here the coefficients of α are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \{\beta, \varepsilon\},$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and λ is a one-form valued in $\bar{A}^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing α as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar{z}_j} d\bar{z}_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar{z}_j}} d\bar{z}_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. Since β appears only linearly in the action, it is a Lagrange multiplier enforcing exactly this equation.

Remark 2.13. This is entirely analogous to BF theory, where B is a Lagrange multiplier enforcing $F_A = 0$, and the phase space is $T^*\mathcal{M}_{\text{flat}}$.

2.5.3. *Phase Space.* Compactify one real direction and put the theory on $\mathbb{R} \times S^1 \times X$ where X is holomorphic symplectic. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan moduli problem for Hamiltonian vector fields on $S^1 \times X$. The perturbative phase space is the derived cotangent stack

$$T^*_{\text{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(X, \mathrm{Ham}(X)), D, \{-,-\}\big).$$

After passing to the underived tangent space at the trivial solution this reduces to the ordinary cotangent bundle of the linearized deformation complex.

2.6. **Closed String Fields and Brane Deformations.** Given a closed string field, we obtain a deformation of the gauge theory on the brane. For example, if the brane lies along x_1 and the closed string field is $dx_1\, z_1^k z_2^l$, we deform the quantum-mechanical action to

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l$$

by adding a matrix-model Hamiltonian. Many matrix Hamiltonians can be engineered by choosing appropriate closed string backgrounds. Construction 2.18 is the algebraic form of this bulk-to-boundary coupling, and Theorem 2.26 below proves that it preserves the classical Hamiltonian bracket.

Remark 2.14. Geometrically, specifying a closed string field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

2.7. **Operators in the Gauge Theory and the Large-N Limit.** The local-operator calculation is performed in the reduced BV model with fields ϕ_1, ϕ_2 and the antifield $\psi = A^*$ for the Lagrange multiplier A. We take GL_N-invariants in characteristic zero, so the Chevalley part of the gauge BRST differential acts trivially on trace functions by cyclicity.

Construction 2.15 (The reduced BV algebra at the insertion point)**.** At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its BRST differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic shadow of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

Lemma 2.16 (De Rham jets)**.** *Positive jets in the topological direction do not contribute to local operator cohomology. Equivalently, every cohomology class has a representative depending only on the values of the fields and antifields at the insertion point.*

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler

homotopy on jet degree. This kills topological-direction jets, not the holomorphic z_i-derivatives used later on the closed side. □

Proposition 2.17 (Stable trace invariants)**.** *Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F,F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The stable invariant algebra*

$$\mathbb{C}[\mathfrak{gl}_\infty^2]^{GL_\infty} := \varprojlim_N \mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$$

with respect to block-restriction maps is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$.

Proof. Procesi–Razmyslov invariant theory [6]razmyslov gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree d. For every $N \geq d$ there are therefore no relations in degree d beyond cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities, and the only surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words.

The same argument applies after adjoining one odd generic matrix ψ and working degree by degree in the exterior variables: invariant functions are generated by graded traces of words in ϕ_1, ϕ_2, ψ, and in the stable range the only relations are graded cyclicity. The Hamiltonian sector used below is the subalgebra generated by the words with zero or one occurrence of ψ. □

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 2.18 (Boundary evaluation of a closed Hamiltonian)**.** Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal

coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\partial_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \operatorname{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Constants give the scalar $c_{0,0}N$ and disappear from the reduced Hamiltonian sector. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0(\mathcal{R}_N^{GL_N}, Q).$$

The first descendant is obtained by inserting one Koszul variable:

$$\partial^{(1)}_{\mathrm{bb},N}(z_1^k z_2^l[1]) = \Psi_{k,l},$$

where $\Psi_{k,l}$ is the symmetrised trace with one ψ, k copies of ϕ_1, and l copies of ϕ_2.

Lemma 2.19 (Adjacent swaps)**.** *In the invariant reduced BV cohomology,*

$$\operatorname{Tr}(u\phi_1\phi_2 v) - \operatorname{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all words u, v.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q \operatorname{Tr}(\psi v u) &= \operatorname{Tr}([\phi_1, \phi_2] v u) \\ &= \operatorname{Tr}(\phi_1\phi_2 v u - \phi_2\phi_1 v u) \\ &= \operatorname{Tr}(u\phi_1\phi_2 v) - \operatorname{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Corollary 2.20 (Matrix evaluation is commutative in cohomology)**.** *If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ depends only on the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$, not on the chosen ordered lift.*

Proof. Adjacent transpositions generate all reorderings of a finite word. Each adjacent transposition of the adjacent letters ϕ_1, ϕ_2 changes the trace by a Q-exact term by Lemma 2.19; cyclicity moves the chosen adjacent pair into the displayed position of that lemma. Summing the resulting exact differences proves the claim. □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\operatorname{Tr}(1) = N$, not to a nontrivial Hamiltonian generator.

Proposition 2.21 (Brane Hamiltonian sector)**.** *In the stable large-N reduced BV model, the classical local operator algebra generated by the non-constant ghost-zero traces and their first Koszul descendants is*

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}[1]\big).$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\mathrm{Tr}(\phi_1^k \phi_2^l)$. The class of $z_1^k z_2^l[1]$ corresponds to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2.

Proof. Proposition 2.17 gives the free commutative algebra on cyclic words. Lemma 2.19 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the ghost-zero cyclic words reduce to the monomials $\mathrm{Tr}(\phi_1^k \phi_2^l)$. The same adjacent-swap argument with one marked occurrence of ψ gives the ghost-(-1) generators; the symmetrised representative is the sum over the multiset consisting of one ψ, k copies of ϕ_1, and l copies of ϕ_2. Products of these single-trace classes give the free graded-commutative multi-trace algebra. By definition of the Hamiltonian sector, single traces containing two or more occurrences of ψ are not part of this algebra. □

2.8. **Operators in the Gravitational Theory.** Let $\mathfrak{h} = \bar{A}$ with Poisson bracket. After choosing the formal Taylor/residue pairing, the cotangent module is identified with a shifted copy of $\bar{A}$, and the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}[1].$$

For $f, g \in \mathfrak{h}$ and $\eta_g = g[1]$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_g] = \eta_{\{f,g\}}, \qquad [\eta_f, \eta_g] = 0.$$

On monomials this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and the same coefficient acts on the shifted generator:

$$[z_1^k z_2^l, \eta_{z_1^m z_2^n}] = (kn - lm) \eta_{z_1^{k+m-1} z_2^{l+n-1}}.$$

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 2.16, together with the Dolbeault Poincare lemma in the formal local model, leaves only holomorphic Taylor coefficients. Thus the linear local coordinates are the

functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

These are dual to the Taylor monomials $z_1^k z_2^l$ up to the factorial normalisation fixed by the chosen pairing. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

Proposition 2.22. *The classical local operators of the formal closed theory are the continuous Chevalley-Eilenberg cochains*

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}^{\vee}[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \bar{A}$. The cotangent field β, after the fixed Taylor/residue pairing, supplies the odd shifted copy $\mathfrak{h}[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}[1]$.

It remains to identify the nonlinear BRST differential. The BV Hamiltonian is

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their Poisson bracket $\{f, g\}$, and the coadjoint action on the cotangent variable is, under the chosen pairing, the adjoint action $[f, \eta_g] = \eta_{\{f,g\}}$. There is no quadratic term with two cotangent inputs, so $[\eta_f, \eta_g] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

2.9. **Koszul Duality.**

Definition 2.23. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C}).$$

We regard commutative Chevalley-Eilenberg cochains as E_1 algebras by restriction.

Proposition 2.24. *Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}[1]$ be the completed graded Lie algebra above, and let*

$$B = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee[-1]\big).$$

Then

$$B^! \simeq \widehat{U(\mathfrak{g})},$$

the completed graded universal enveloping algebra.

Proof. Let

$$C^{\mathrm{CE}}_*(\mathfrak{g}) = \mathbf{S}^c(\mathfrak{g}[1])$$

be the Chevalley-Eilenberg coalgebra, with coderivation determined by the bracket of $\mathfrak{g}$. By definition

$$B = \mathrm{Hom}_{\mathrm{cont}}\big(C^{\mathrm{CE}}_*(\mathfrak{g}), \mathbb{C}\big).$$

The augmentation $B \to \mathbb{C}$ has the standard bar resolution, and continuous duality identifies

$$\mathrm{RHom}_B(\mathbb{C}, \mathbb{C}) \simeq \Omega\, C^{\mathrm{CE}}_*(\mathfrak{g}),$$

the completed cobar algebra of the CE coalgebra. The Lie bar-cobar comparison [8]loday-vallette identifies this cobar algebra with the completed associative algebra generated by $\mathfrak{g}$ modulo the graded Lie relations; this is $\widehat{U(\mathfrak{g})}$.

Thus, for the generators $f \in \mathfrak{h}$ and $\eta_g \in \mathfrak{h}[1]$, the completed enveloping algebra has relations

$$\begin{aligned} fg - gf &= \{f, g\}, \\ f\eta_g - \eta_g f &= \eta_{\{f,g\}}, \\ \eta_f \eta_g + \eta_g \eta_f &= 0. \end{aligned}$$

□

For comparison with the classical brane algebra introduce the Rees enveloping algebra $U_\hbar(\mathfrak{g})$ with relations

$$xy - (-1)^{|x||y|} yx = \hbar[x, y].$$

At $\hbar = 0$ one obtains

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \mathbf{S}(\mathfrak{g}) = \mathbf{S}\big(\bar{A} \oplus \bar{A}[1]\big),$$

which is the stable classical brane operator algebra of the Hamiltonian sector in Proposition 2.21. Thus the free, abelian closed theory is Koszul dual directly to $\mathbf{S}(\bar{A} \oplus \bar{A}[1])$, while the interacting Hamiltonian bracket deforms this algebra to the completed enveloping algebra via PBW.

2.9.1. *First-order corrections ($\hbar^1$).* Under the Koszul duality correspondence we have

$$z_1^k z_2^l \leftrightarrow \mathrm{Tr}(\phi_1^k \phi_2^l).$$

The Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\, z_1^m z_2^n\} = (kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}.$$

The matching bracket on the open side is not an A-propagator computation: A is a Lagrange multiplier. It is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

Proposition 2.25. *In the stable reduced cohomology,*

$$\{\mathrm{Tr}(\phi_1^k \phi_2^l), \mathrm{Tr}(\phi_1^m \phi_2^n)\} = (kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1}).$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a} \phi_2^l \phi_1^a,$$

$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b} \phi_1^k \phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either sum has multidegree $(k+m-1, l+n-1)$. Lemma 2.19 identifies all cyclic words with this multidegree in reduced BRST cohomology. The coefficient is therefore $kn - lm$, and the common class is $\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})$. □

Theorem 2.26 (Classical bulk-to-boundary Hamiltonian map)**.** *In the stable reduced Hamiltonian sector, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 2.18 stabilize to a linear map*

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^0(\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2).$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra to the open trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\mathbb{C}^2}\big).$$

Moreover, a closed background of the form $a(t) f\, dt$ along the brane deforms the open action by the boundary term

$$\int_{\mathbb{R}} a(t)\, \partial_{\mathrm{bb}}(f)\, dt.$$

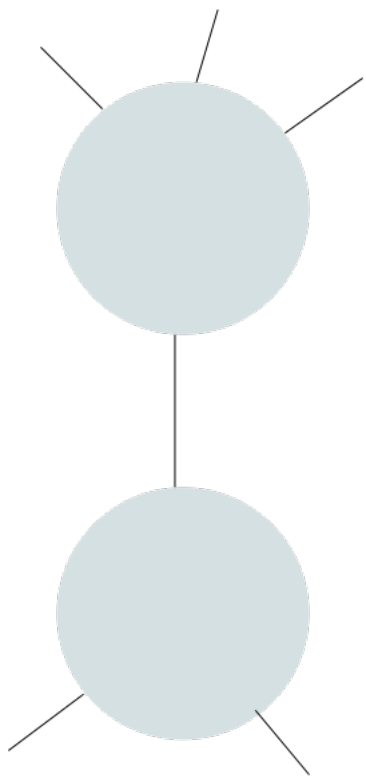
Proof. The map is well-defined on $\bar{A}$ by Corollary 2.20; constants map to the scalar N and are removed in the reduced sector. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 2.25 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = (kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1}).$$

The closed Poisson bracket is

$$\{f, g\}_{\mathbb{C}^2} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. The displayed deformation of the action is exactly the boundary evaluation construction integrated against the one-form coefficient $a(t)dt$ of the closed background. □



The picture records this single canonical contraction.

Proof of Theorem 1.4. The open Hamiltonian-sector algebra is obtained from the product of the exact A-model self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ and the affine Koszul calculation for $\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m})$. Lemma 2.4 identifies the ghost-zero reduction with the gauged first-order matrix model

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Lemma 2.16 removes positive topological jets, and Lemma 2.19 identifies cyclic words with the same commutative monomial. Proposition 2.17 then gives Proposition 2.21:

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}} \cong \mathbf{S}(\bar{A} \oplus \bar{A}[1]).$$

On the closed side, the divergence computation and Lemma 2.9 identify the local Hamiltonian modes with $\bar{A}$. The cotangent BF field supplies the shifted adjoint copy, so the local Lie algebra is $\mathfrak{g} = \bar{A} \ltimes \bar{A}[1]$. Proposition 2.22 computes local observables as $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$.

Proposition 2.24 gives the Koszul dual $\widehat{U(\mathfrak{g})}$, and the Rees/PBW specialization gives $\mathbf{S}(\mathfrak{g}) = \mathbf{S}(\bar{A} \oplus \bar{A}[1])$ at $\hbar = 0$. Finally Theorem 2.26 constructs the map from closed Hamiltonians to brane Hamiltonians and shows that it preserves the Poisson bracket. On monomials this gives the coefficient $(kn - lm)$ on both sides. □

2.9.2. *Second-order corrections ($\hbar^2$).* We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus there is no $\hbar^2$ term in this quantization convention.

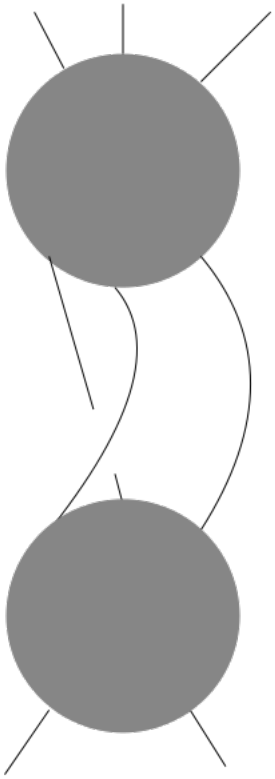
2.9.3. *Third-order corrections ($\hbar^3$).* For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient is completely explicit:

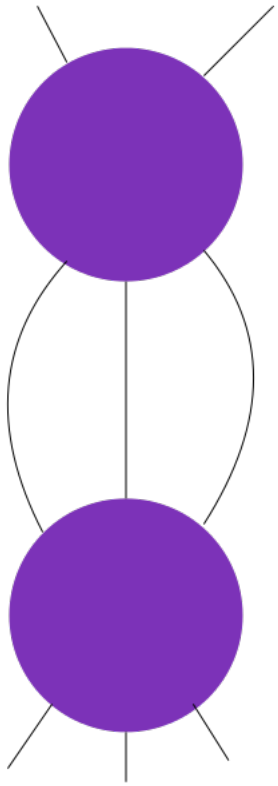
$$P^3(f, g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a} (l)_a (m)_a (n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f, g]_* = \hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5).$$

The script `scripts/check_moyal_coefficients.py` checks this first-order coefficient, the vanishing of the second-order commutator coefficient, the displayed third-order formula, and the star-commutator normalisation on every monomial pair with exponents between 0 and 6, using exact rational arithmetic.





The displayed ribbon graphs are only graph-topology reminders. To turn them into a theorem one still has to specify the closed-string gauge fixing, bulk-to-boundary map, propagator on the noncompact surface, trace normalisation, and large-N weighting. Those data are not part of the present calculation, so no multi-trace third-order coefficient is asserted here.

Remark 2.27. In characteristic zero, passing to underived GL_N-invariants is exact on rational representations. This is the convention used in the reduced brane-operator computation above. Keeping the full derived quotient would retain the Chevalley ghosts and produces a larger BRST complex; that refinement is not used in the stable comparison made here.

Remark 2.28. On the gravity side we did not choose a brane: we worked with string states. On the gauge theory side we chose a brane of the form a point cross a Lagrangian. By choosing a different B-brane, we move away from the structure of an associative algebra to a more general factorization algebra. Koszul duality should

extend to this setting. For example, with two topological directions (instead of one) we would obtain an E_2 algebra and hence E_2 Koszul duality.

Remark 2.29. One could also study D_0 branes on $\mathbb{R}^2 \times \mathbb{C}^4$ from this point of view.

[1] M. Bershadsky, S. Cecotti, H. Ooguri, and C. Vafa, *Kodaira-Spencer theory of gravity and exact results for quantum string amplitudes*, Comm. Math. Phys. **165** (1994), 311–428.
[2] J.-L. Loday and D. Quillen, *Cyclic homology and the Lie algebra homology of matrices*, Comment. Math. Helv. **59** (1984), 565–591.
[3] B. L. Tsygan, *Homology of matrix Lie algebras over rings and the Hochschild homology*, Uspekhi Mat. Nauk **38** (1983), 217–218.
[4] E. Witten, *Chern-Simons gauge theory as a string theory*, The Floer Memorial Volume, Progr. Math., vol. 133, Birkhäuser, Basel, 1995, pp. 637–678.
[5] M. Alexandrov, M. Kontsevich, A. Schwarz, and O. Zaboronsky, *The geometry of the master equation and topological quantum field theory*, Internat. J. Modern Phys. A **12** (1997), no. 7, 1405–1430.
[6] C. Procesi, *The invariant theory of $n \times n$ matrices*, Advances in Math. **19** (1976), no. 3, 306–381.
[7] Yu. P. Razmyslov, *Trace identities of full matrix algebras over a field of characteristic zero*, Math. USSR Izv. **8** (1974), no. 4, 727–760.
[8] S. B. Priddy, *Koszul resolutions*, Trans. Amer. Math. Soc. **152** (1970), 39–60.
[9] Jean-Louis Loday and Bruno Vallette, *Algebraic Operads*, Grundlehren der Mathematischen Wissenschaften, vol. 346, Springer, Heidelberg, 2012.

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